

The recession cone of convex sets

Kent Andersen

In this note we consider unbounded directions for convex sets. The main question is: For a convex set $C \subseteq \mathbb{R}^n$, along which directions is it possible to move infinitely far away from points in C and remain in C , or equivalently, for which $d \in \mathbb{R}^n$ does one have $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq C$ for all $x \in C$? The directions along which this is possible are called *recession directions*.

Definition 1 (*Recession directions and recession cones for convex sets*)
A vector $d \in \mathbb{R}^n$ is called a recession direction for a convex set $C \subseteq \mathbb{R}^n$ iff $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq C$ holds for all $x \in C$. The set of recession directions for C is called the recession cone of C , and it is denoted $\text{rec}(C)$. ■

We next characterize the recession cone of a polyhedron. For a polyhedron $P \subseteq \mathbb{R}^n$, and a point $x \in P$, the following lemma shows which directions $d \in \mathbb{R}^n$ satisfy $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P$.

Lemma 1 (*Recession directions for polyhedra*)
Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron with $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. A point $x \in P$ and a direction $d \in \mathbb{R}^n$ satisfies $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P \iff Ad \leq 0_m$. ■

Proof: " $\Rightarrow$ ": Suppose $x + \alpha \cdot d \in P$ for all $\alpha \geq 0$, i.e., suppose $A(x + \alpha \cdot d) \leq b$ for all $\alpha \geq 0$. This implies $\alpha \cdot (Ad) \leq b - Ax$ for all $\alpha \geq 0$. If one had $(Ad)_i > 0$ for some coordinate $i \in \{1, \dots, m\}$, then one would have $\alpha \cdot (Ad)_i > b_i - (Ax)_i$ for $\alpha > 0$ sufficiently large, which would be a contradiction. Hence $Ad \leq 0_m$.

" $\Leftarrow$ ": Now suppose $Ad \leq 0_m$. Then $A(x + \alpha \cdot d) \leq b + \alpha \cdot 0_m = b$ for all $\alpha \geq 0$, which implies $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P$. ■

Observe that the condition in Lemma 1 does not depend on the starting point: If $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P$ holds for *some* $x \in P$, then $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P$ holds for *all* $x \in P$.

Observation 1 (*Independence of recession directions from the starting point*)
For any polyhedron $P \subseteq \mathbb{R}^n$ and direction $d \in \mathbb{R}^n$, if $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P$ holds for some $x \in P$, then $\{x + \alpha \cdot d : \alpha \geq 0\} \subseteq P$ holds for all $x \in P$. ■

It follows that Lemma 1 gives a description of the recession cone $\text{rec}(P)$:

Theorem 1 (*The recession cone for a polyhedron*)

The recession cone of a polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ with $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ is given by $\text{rec}(P) = \{x \in \mathbb{R}^n : Ax \leq 0_m\}$. ■

We now characterize the recession cone of a finitely generated set. We therefore fix a finitely generated set $Q = \text{conv}(V) + \text{cone}(E)$ in the remainder, where $V, E \subseteq \mathbb{R}^n$ are finite sets. We will show $\text{rec}(Q) = \text{cone}(E)$.

Theorem 2 (*The recession cone of a finitely generated set*)

The recession cone of $Q = \text{conv}(V) + \text{cone}(E)$ is given by $\text{rec}(Q) = \text{cone}(E)$. ■

This result is proven by combining Theorem 1 with the fact that finitely generated cones are polyhedral cones (proved in note 1.(D)):

Fact 1 *Finitely generated cones are polyhedral cones.* ■

Consider the so-called *cone extension* of Q , which is the following finitely generated set:

$$C_f(V, E) := \text{cone}\left(\left\{\begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V\right\}\right) + \text{cone}\left(\left\{\begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E\right\}\right).$$

Observe that one has $\begin{pmatrix} x \\ 1 \end{pmatrix} \in C_f(V, E) \iff x \in \text{conv}(V) + \text{cone}(E)$ and $\begin{pmatrix} x \\ 0 \end{pmatrix} \in C_f(V, E) \iff x \in \text{cone}(E)$. Since $C_f(V, E)$ is a polyhedral cone, Fact 1 shows one may write:

$$C_f(V, E) = \left\{\begin{pmatrix} x \\ \lambda \end{pmatrix} \in \mathbb{R}^n \times \mathbb{R} : Dx - d\lambda \leq 0_p\right\},$$

where $D \in \mathbb{R}^{p \times n}$ is a matrix and $d \in \mathbb{R}^p$. It follows that:

$$\begin{aligned} Q &= \{x \in \mathbb{R}^n : \begin{pmatrix} x \\ 1 \end{pmatrix} \in C_f(V, E)\} = \{x \in \mathbb{R}^n : Dx \leq d\}, \text{ and} \\ \text{cone}(E) &= \{x \in \mathbb{R}^n : \begin{pmatrix} x \\ 0 \end{pmatrix} \in C_f(V, E)\} = \{x \in \mathbb{R}^n : Dx \leq 0_p\}. \end{aligned}$$

Applying Theorem 1 shows $\text{rec}(Q) = \{x \in \mathbb{R}^n : Dx \leq 0_p\} = \text{cone}(E)$. This completes the proof of Theorem 2.